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Design Ideas for Oracle and Diffusion Operator Circuits

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1 Introduction

Grover's algorithm is a quantum search algorithm used to identify a target element in an unstructured set of N items. What makes Grover's algorithm faster than classical search is that it uses quantum superposition and interference to reduce the number of oracle queries. Thus, a classical search requires O(N) calls to a classical oracle, while Grover's algorithm requires O($\sqrt{N}$) queries to a quantum oracle [1]. Grover's algorithm involves querying an oracle to mark the target state and performing a specific series of gate operations, which together form the diffusion operator to increase the probability of finding a specific target state. In this report, we provide a detailed explanation of how the oracle and diffusion operator quantum circuits are designed in the QPanda3 SDK to find a bit string of $|1010\rangle$.

2 Design Idea and Implementation of Grover's Oracle Circuit

To find the bit string $|1010\rangle$, 4 qubits are first initialized using a set of Hadamard gates, which creates a superposition of all possible states within our search space N before the oracle is applied to them. The function of the oracle is to mark the target state by flipping its phase from positive to negative so its probability can be amplified in subsequent operations. In implementing the oracle, this is achieved by applying a multi-controlled Z gate (MCZ) to the initialized qubits. Moreover, since the MCZ gate only works on the $|1111\rangle$ state to achieve a phase flip, the construction of the oracle will depend on the type of state we are trying to find, which will determine whether or not a set of X gates would be sandwiched between the MCZ gate to achieve the phase flipping effect for the targeted state. The MCZ gate is, therefore, not implemented alone in the case of finding the bit string $|1010\rangle$. After initialization, the MCZ gate needs to be sandwiched by two pairs of X gates applied on the first and third qubits - q_0 and q_2 - according to qPanda3's little endian notation $|q_3q_2q_1q_0\rangle$. The first pair of X gates ensures that the state of the target qubit is transformed into the $|1111\rangle$ state before the MCZ gate is applied, while the second pair of X gates is implemented to uncompute the transformed quantum state back to its original form, with the phase flip effect implemented only on the target state. Mathematically, the transformation from initialization to implementation of the oracle is as follows:

$$|\phi_0\rangle = |0000\rangle \tag{1}$$

$$|\psi_0\rangle = H^{\otimes 4}|0000\rangle = \frac{1}{\sqrt{16}}\sum_{x=0}^{15}|x\rangle = \frac{1}{4}\sum_{x=0}^{15}|x\rangle \tag{2}$$

$$|\psi_1\rangle = (I \otimes X \otimes I \otimes X)\ |\psi_0\rangle = \frac{1}{4}\sum_{x=0}^{15}|x\rangle \tag{3}$$

$$|\psi_2\rangle = \mathrm{MCZ}\ |\psi_1\rangle = \frac{1}{4}\left(\sum_{x=0}^{14}|x\rangle\ -\ |1111\rangle\right) \tag{4}$$

$$|\psi_3\rangle = (I \otimes X \otimes I \otimes X)\ |\psi_2\rangle = \frac{1}{4}\left(\sum_{\substack{x=0\\x\neq 10}}^{15}|x\rangle\ -\ |1010\rangle\right) \tag{5}$$

Hence, we get the final oracle operator for the target state $|1010\rangle$ and subsequent circuit implementation as follows:

$$U_\omega\ =\ (I \otimes X \otimes I \otimes X)\,\mathrm{MCZ}\,(I \otimes X \otimes I \otimes X)\ =\ I - 2\,|1010\rangle\langle 1010| \tag{6}$$

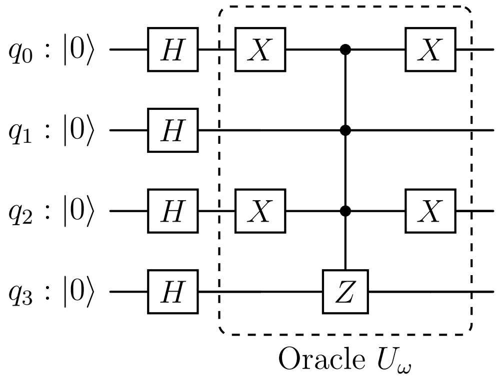

Figure 1: Initialization of qubits together with Oracle circuit to mark the target state $|1010\rangle$.

3 Design Idea and Implementation of Diffusion Operator Circuit

Once the state is marked, the diffusion operator is applied, which reflects all amplitudes about the mean probability amplitude, thereby increasing the amplitude of the marked state while reducing that of unmarked states. Unlike the oracle, the diffusion operator is applied uniformly and does not depend on the type of state being sought. The gate implementation for the diffusion operator we designed, which correctly satisfies the general equation for the diffusion operator, is given by:

$$D \;=\; H^{\otimes 4}\, X^{\otimes 4}\, \mathrm{MCZ}\, X^{\otimes 4}\, H^{\otimes 4} \;=\; 2\,|\psi_0\rangle\,\langle\psi_0| - I, \qquad |\psi_0\rangle = H^{\otimes 4}\,|0000\rangle = \frac{1}{4}\sum_{x=0}^{15}|x\rangle \tag{7}$$

From (7), we can see that Hadamard gates are applied first to all the qubits, followed by a set of X gates, an MCZ gate, another set of X gates, before finally applying the last layer of H gates. The Hadamard gates are applied first to transform the axis of reflection effect of the inner gate layers of the diffusion operator about the uniform superposition state. When we apply the next layer of the diffusion operator, only the $|0000\rangle$ state then gets mapped onto the $|1111\rangle$ state, ensuring that the MCZ gate flips the phase of the new $|1111\rangle$ state such that after applying the next layer of X gates, the phase on the $|0000\rangle$ state is flipped. Applying the last layer of H gates produces an inversion of the amplitude of the target state about the mean amplitude, causing a constructive interference on the target state and a uniform suppression of the probability amplitudes of the non-target states, effectively amplifying the probability of measuring the target state.

The inner layer produces a reflection about the $|0000\rangle$ state which when compared to the standard diffusion operator from the original grover's paper [reference], differ by a phase which can be ignored during measurement. Thus, we get the inner layer implementation to be:

$$\mathrm{MCZ} \;=\; I - 2\,|1111\rangle\langle 1111|\,. \tag{8}$$

$$\begin{aligned} X^{\otimes 4}\,\mathrm{MCZ}\,X^{\otimes 4} &= X^{\otimes 4}\big(I - 2\,|1111\rangle\langle 1111|\big)X^{\otimes 4} && (9)\\ &= X^{\otimes 4} I X^{\otimes 4} \;-\; 2\big(X^{\otimes 4}\,|1111\rangle\big)\big(\langle 1111|\,X^{\otimes 4}\big) && (10)\\ &= I - 2\,|0000\rangle\langle 0000|\,. && (11) \end{aligned}$$

$$X^{\otimes 4}\,\mathrm{MCZ}\,X^{\otimes 4} \;=\; -\big(2\,|0000\rangle\langle 0000| - I\big), \tag{12}$$

Ignoring global phase, we get the standard grover diffusion operator:

$$D \;=\; H^{\otimes 4}\,X^{\otimes 4}\,\mathrm{MCZ}\,X^{\otimes 4}\,H^{\otimes 4} \;=\; I - 2\,|\psi_0\rangle\langle\psi_0| \;\equiv\; 2\,|\psi_0\rangle\langle\psi_0| - I. \tag{13}$$

The diffusion operator circuit design is therefore given below:

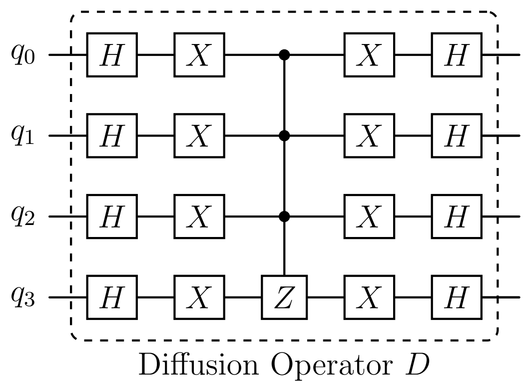

Figure 2: Diffusion operator circuit.

4 References

[1] Lov K. Grover. "A fast quantum mechanical algorithm for database search". In: *Proceedings of the Twenty-Eighth Annual ACM Symposium on Theory of Computing*. STOC '96. Philadelphia, Pennsylvania, USA: Association for Computing Machinery, 1996, pp. 212–219. ISBN: 0897917855. DOI: 10.1145/237814.237866. URL: https://doi.org/10.1145/237814.237866.